STEERING—STEERING SHUDDER—FELT WHEN TURNING—DRIVING AT SLOW SPEEDS OR AT STOP—4X2 AND 4X4 SPORT TRAC VEHICLES ONLY	Article No. 01-25-1

FORD: 2001-2002 EXPLORER SPORT TRAC

This article is being republished in its entirety to update the Part Numbers and the vehicle coverage.

ISSUE

Some 2001-2002 Explorer Sport Trac 4X2 and 4X4 vehicles may exhibit a steering wheel shudder during low-speed turning maneuvers. This may be due to the interaction between the tire and the road surface, which is amplified by the steering system. The condition will vary according to ambient temperature, road surface friction, and steering wheel turning speed.

ACTION

Replace the power steering pump and cooler with redesigned service parts. Install a no-hole check valve in the steering gear. Perform a thorough evacuation of air from the power steering system after replacement/installation of parts and reprogram the Powertrain Control Module (PCM) to the latest level calibration. THE EVACUATION MUST BE PERFORMED AS DESCRIBED BELOW OR THE SHUDDER MAY STILL EXIST AFTER THE REPAIR. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

NOTE

THE STEERING SHUDDER WILL NOT BE CORRECTED UNLESS THE SYSTEM IS PURGED OF AIR WITH A VACUUM AT THE RESERVOIR. CORRECTLY FOLLOWING THE EVACUATION PROCEDURE DETAILED BELOW WILL REDUCE THE AMOUNT OF TIME NEEDED TO CORRECT THE SHUDDER. THE SHUDDER MAY STILL BE PRESENT AFTER REPLACEMENT OF THE PARTS IF THE EVACUATION PROCEDURE IS NOT PROPERLY FOLLOWED.

NOTE

PRIOR TO PERFORMING THIS PROCEDURE, PERFORM THE DIAGNOSIS AND TESTING DESCRIBED IN THE APPROPRIATE WORKSHOP MANUAL, SECTION 211-00 TO CONFIRM THAT ALL OTHER SYSTEM ISSUES HAVE BEEN IDENTIFIED AND RESOLVED.

1. Replace the Power Steering Pump with a new high displacement pump (1L2Z-3A674-GA). BEFORE REPLACEMENT, CLEAN ALL AREAS WHERE DIRT INGESTION COULD OCCUR. FAILURE TO DO SO MAY RESULT IN SYSTEM CONTAMINATION AND NOISE. Refer to the Workshop Manual, Section 211-02 for replacement instructions.

NOTE

DO NOT FILL, PURGE, OR LEAK CHECK THE SYSTEM AT THIS POINT SINCE THE STEERING GEAR CHECK VALVE AND COOLER MUST ALSO BE REPLACED.

2. Replace the Cooler Assembly with service part F77Z-3D746-GA. (This new cooler has a larger fin pack to maintain lower fluid temperatures). BEFORE REPLACEMENT, CLEAN ALL AREAS WHERE DIRT INGESTION COULD OCCUR. FAILURE TO DO SO MAY RESULT IN SYSTEM CONTAMINATION AND NOISE. The existing cooler return line (hose attached to the gear outlet port) will be reused to ease installation. Attach a clamp to the cooler return line to reduce fluid spillage during removal of the cooler. Disconnect the existing cooler fin pack from the cooler return line by releasing the constant tension clamp. Remove the return line on the new cooler to be installed and connect this new fin pack assembly to the existing return line. Resecure the constant tension clamp. Refer to the Workshop Manual, Section 211-02 for complete replacement instructions.

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3. Disconnect the power steering pressure line from the gear pressure port to allow replacement of the check valve. DO NOT ATTACH A CLAMP TO THE POWER STEERING PRESSURE HOSE AS IT MAY CAUSE DAMAGE TO THE TUNING CABLE INSIDE THE HOSE. Attach a clamp to the COOLER RETURN LINE to reduce fluid flow out of the gear during removal of the pressure line. Loosen the nut securing the power steering pressure hose to the gear inlet and detach it from the gear. REPLACE THE O-RING (388897-S) ON THE POWER STEERING PRESSURE HOSE NUT. USE ROTUNDA TOOL 211-D-027 OR EQUIVALENT TO SIZE THE O-RING. ANY OTHER METHOD OF INSTALLATION WILL DEFORM IT, MAKING IT INEFFECTIVE.

4. Replace the check valve located in the steering gear pressure side inlet port with a No-Hole Check Valve (E6DZ-3N603-A). This new check valve has a spring-loaded closure allowing flow in only one direction, without the bleeder hole that is present in the current check valve. CLEAN ALL AREAS WHERE DIRT INGESTION COULD OCCUR. FAILURE TO DO SO MAY RESULT IN SYSTEM CONTAMINATION AND NOISE. Using a pick tool, remove the clip retaining the check valve in the gear inlet. Then, using the pick, remove the check valve from the port. Install the no-hole check valve WITH THE OPEN SIDE DOWNWARD into the pressure side inlet of gear (Figures 1 and 2). Reattach the power steering pressure hose to the inlet of the steering gear.

5. Fill the power steering reservoir with MERCON® Multi-Purpose (ATF) Transmission Fluid (XT-2-QDX) or equivalent.

6. WITH THE ENGINE OFF, apply 25 inches of vacuum to the reservoir opening. Start the engine and let idle for 1 minute. DO NOT TURN THE STEERING WHEEL. If power steering reservoir becomes low, remove the vacuum and refill.

7. WITH THE ENGINE RUNNING and NO VACUUM APPLIED TO THE SYSTEM, turn the steering wheel to the left and then to the right end-of-travel stops, KNOCKING AGAINST EACH STOP SEVERAL TIMES to purge remaining air from the steering system. Air bubbles should be seen rising from the reservoir. Use of a shop light behind the reservoir will facilitate monitoring of the release of air. Repeat this procedure four times in each direction.

NOTE
AT NO TIME DURING THIS PROCEDURE SHOULD THE STEERING WHEEL BE HELD ON THE STOPS OR THE POWER STEERING PUMP COULD BE DAMAGED.

8. Recheck the fluid level and, if necessary, refill.

9. Test drive the vehicle and evaluate. Perform static parking maneuvers, IDEALLY ON PAINTED CONCRETE OR SEALED ASPHALT, to evaluate the vehicle for steering shudder and any other steering concerns.

10. If steering shudder is still present, or power steering pump "whine" is occurring, a more thorough air evacuation may be needed. With 25 inches of vacuum applied, start the engine and stand outside the vehicle. Turn the steering wheel from lock-to-lock several times while watching the reservoir for bubbles (using a shop light behind the reservoir), and listen for any sign of power steering noise. When bubbles are no longer rising out of the reservoir, and any power steering system noise has ceased, shut off the engine.

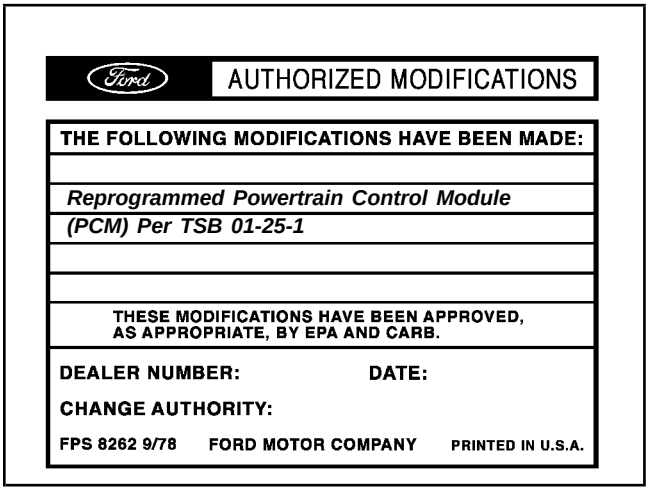
NOTE
THIS SHOULD NOT BE OVERDONE OR IT COULD DAMAGE THE PUMP.

11. Reprogram the PCM to the latest level.

12. Test drive to verify repair.

PCM CALIBRATION INFORMATION						
Application	Old Part Number (-12A650-)	Tear Tag	New Part Number (-12A650-)	Old Calibration	New Calibration	NGS/WDS Qualifier
2001-2002 Sport Trac	1L2F-AAC	JXV3	1L2Z-AAD	1S11A40505	1S11A40506	WDS B16.2 Release or Later

Obtain an Authorized Modifications Decal (STAT 5941-2 - obtainable through the PDC) and list the date, dealer number, and summary of modifications performed. Select a prominent place adjacent to the Vehicle Emission Control Information Decal suitable for installing the Authorized Modifications Decal. Clean the area, install the decal, and cover it with a clear plastic decal shield.



PART NUMBER	PART NAME
1L2Z-3A674-GA	Power Steering Pump
F77Z-3D746-GA	Power Steering Fluid Cooler
E6DZ-3N603-A	No-Hole Check Valve
388897-S	Teflon O-Ring

OTHER APPLICABLE ARTICLES: NONE

SUPERSEDES: 01-10-4

WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

OPERATION	DESCRIPTION	TIME
012501A	Replace Power Steering Pump, Fluid Cooler, And Install No-Hole Check Valve In The Steering Gear Then Reprogram Powertrain Control Module (Includes Time To Evacuate Air From The System)	1.9 Hrs.
012501B	Replace Power Steering Pump, Fluid Cooler, And Install No-Hole Check Valve In The Steering Gear Then Reprogram Powertrain Control Module (Includes Time To Evacuate Air From The System And A Second Time When Needed)	2.4 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
3504	42

OASIS CODES: 303000, 703000, 703200, 703300, 703400

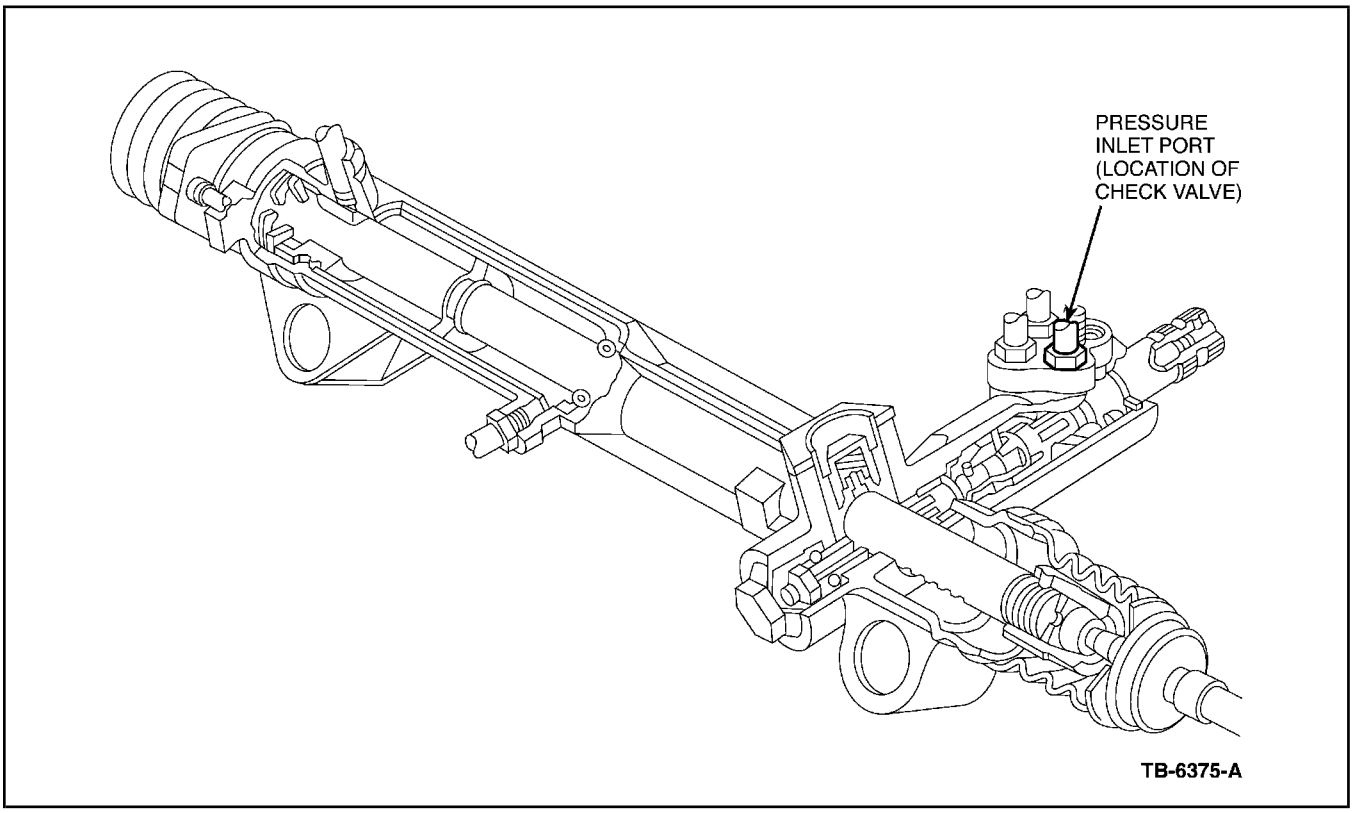


Figure 1 - Article 01-25-1

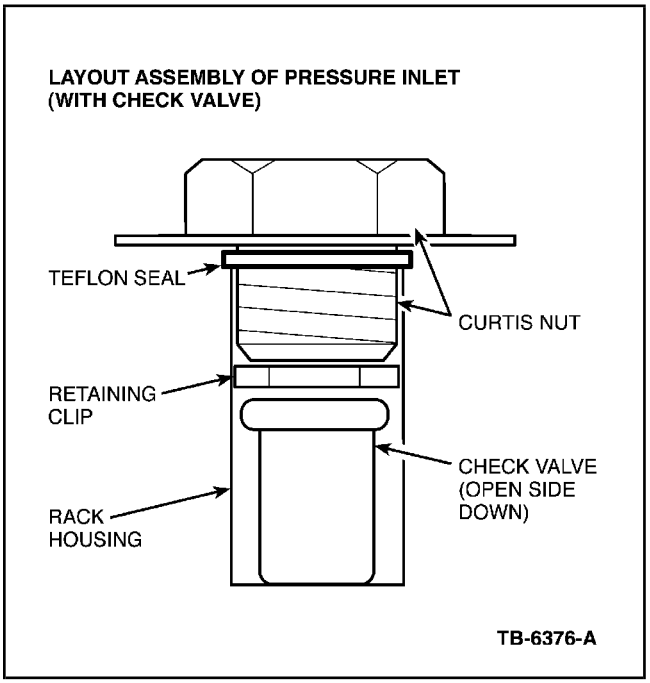


Figure 2 - Article 01-25-1